

FRONT MATTER

Comprehensive Profiles of Human Cingulate Gyrus Effective Networks

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30 **Abstract**

31 The human cingulate gyrus supports diverse functions, yet its effective networks lack
32 standardized, network-level models. We mapped the effective networks of the anterior, mid, and
33 posterior cingulate in 15 resection-free patients using single-pulse electrical stimulation and
34 multimodal analysis of cortico-cortical evoked potentials (CCEPs). A morphology-based CCEP
35 coherence measure complemented conventional amplitude-based measures, revealing connections
36 undetected by amplitude alone. We also identified cortical hub regions with near equal
37 connectivity across cingulate subregions. Spatial profiles of these effective networks revealed an
38 anterior-posterior connectivity gradient consistent with known cingulum white-matter
39 organization. Hierarchical clustering indicated the midcingulate network exhibits the most
40 stereotyped CCEP morphology across regions. Finally, a random forest classification model
41 trained on CCEP features distinguished cingulate subregion networks with 75% accuracy and
42 revealed that commonly used assumption-based features had the lowest predictive strength. These
43 results motivate multimodal, data-driven approaches to modeling effective connectivity and
44 provide detailed, network-level models of the human cingulum.

46 **Teaser**

47 Intracranial stimulation of the cingulate cortex reveals hub regions and spatial gradients, as well
48 as which measures are most important to characterize effective networks.

50 **MAIN TEXT**

51 **Introduction**

52 In humans, the cingulate gyrus (Cg) is widely recognized as a central locus involved in a
53 broad range of processes. Forming an arch-shaped convolution directly dorsal to the corpus
54 callosum, the Cg has historically been defined by its functional, anatomical, connective, and
55 cytoarchitecturally distinct subdivisions (1–9). The most formal model of the Cg groups these
56 subdivisions into four regions: the anterior cingulate cortex (ACC), midcingulate cortex (MCC),
57 posterior cingulate cortex (PCC), and retrosplenial cortex (3, 6).

58 The ACC is one of the most widely studied cortical regions and is implicated in cognitive
59 control, autonomic control, social cognition, decision-making, prediction error, pain, and emotion
60 (10–16). The ACC’s diverse functional profile affords an equally intricate repertoire of functional
61 and structural connections. The most well-documented connections are with limbic, frontal,
62 orbitofrontal, parietal, thalamic, cingulate, and midbrain structures (2, 4, 8, 17). The MCC has
63 two functionally and connectively distinct divisions consisting of its anterior and posterior

While CCEP analysis has the potential to reveal invaluable information on the dynamics and function of neural systems, the lack of a formally established methodology and analysis standards limits the utility and interpretability of effective connectivity characterization (29, 32). This is especially true for evaluating the effective networks of the Cg, as its functional complexity and highly integrated nature likely follow intricate effective connections throughout the brain. Current studies of effective connectivity, even those evaluating the networks of the Cg, use highly variable stimulation parameters, drastically affecting CCEP characteristics, or include data from patients with major cortical resections, which alters tractography and cortical physiology (27, 33, 34). Analytical methods also have an effect on the statistical content and interpretability of CCEPs. Therefore, the choice of analytical method directly determines how the effective connections between regions are characterized and defined. For example, studies of effective connectivity often use CCEP magnitude or rely on pre-defined assumptions about CCEP shape, thus limiting the utility and depth of network-wide characterization (28, 29). Further, due to the high dimensionality of data required to profile entire effective networks, studies of effective connectivity utilizing more rigorous analytical approaches often focus on subsets of networks, especially in work investigating hypothesis-preselected circuits. While these methods are critical for providing deep insight into the neural mechanisms of the connections comprising networks, this level of detail makes it difficult to profile entire effective networks and compare them (29, 32, 35). Furthermore, understanding effective networks, especially those of the human Cg, has direct clinical relevance as subregions are targeted for interventions to treat a diverse set of neuropsychiatric disorders, including depression, anxiety, and pain (36–38).

The present study presents a comprehensive comparison between the divergent profiles of the human ACC, MCC, and PCC (including retrosplenial cortex) effective networks acquired through direct SPES. Data were acquired using constant SPES parameters from patients without any resections. Cg subregion profiles were modeled across a breadth of features extracted from CCEPs, including magnitude, morphology, morphological diversity, inter-regional similarity, spatial patterning, temporal dynamics, broadband gamma content, assumption-free waveform features, and assumption-based waveform features. Our study provides a network-level profile of the characteristics, similarities, and differences between the Cg effective networks modeled across several metrics. Further, through a machine-learning approach, we provide direct, quantitative evidence on the utility of effective connectivity metrics used to model effective networks, specifically by characterizing their predictive strength in classifying Cg networks. In aggregate, this work provides a comprehensive foundation for classifying the complex interactions that the Cg has with the rest of the brain.

Results

Patient Demographics, SEEG Coverage, and Stimulation Locations

Our models of Cg effective connectivity were derived from invasive electrophysiological data obtained from 15 patients with intractable epilepsy undergoing invasive SEEG monitoring (Age: mean $\pm$ SD, 33.3 ± 11.85). Details, including patient duration of invasive monitoring, and clinical details, can be found in Supplemental Table 1. Experimental design is shown in Figure 1a.

The number of channels within each major brain region across patients in our study population is shown in Figure 1b. There was a median of 156 contacts $\pm$ 64.5 MAD across major brain regions, with the temporal lobe containing 1060 channels, the frontal lobe containing 484 channels, and the orbitofrontal cortex containing 223 channels. The number of contacts within the cingulate cortex (167), somato-motor cortices (198), hippocampus (145), amygdala (94), and insula (212) were within the MAD across all regions. The operculum (88), parietal lobe (80), and thalamus (16) were the least sampled regions.

Figure 1c shows a breakdown of the stimulation locations within the Cg. Within our study population, 40% of stimulations were delivered to the left Cg, with 18.4% of stimulations occurring at the left ACC, 14.8% at the left MCC, and 6.4% at the left PCC, and 60% were delivered to the right Cg, with 26.4% of stimulations occurring at the right ACC, 24.9% at the right MCC, and 8.8% at the right PCC.

Figure 1d shows a set of exemplar CCEPs recorded from different brain regions in response to stimulation of the ACC, MCC, or PCC. The morphological characteristics of the CCEPs vary depending on where the response was recorded and which subregion was stimulated.

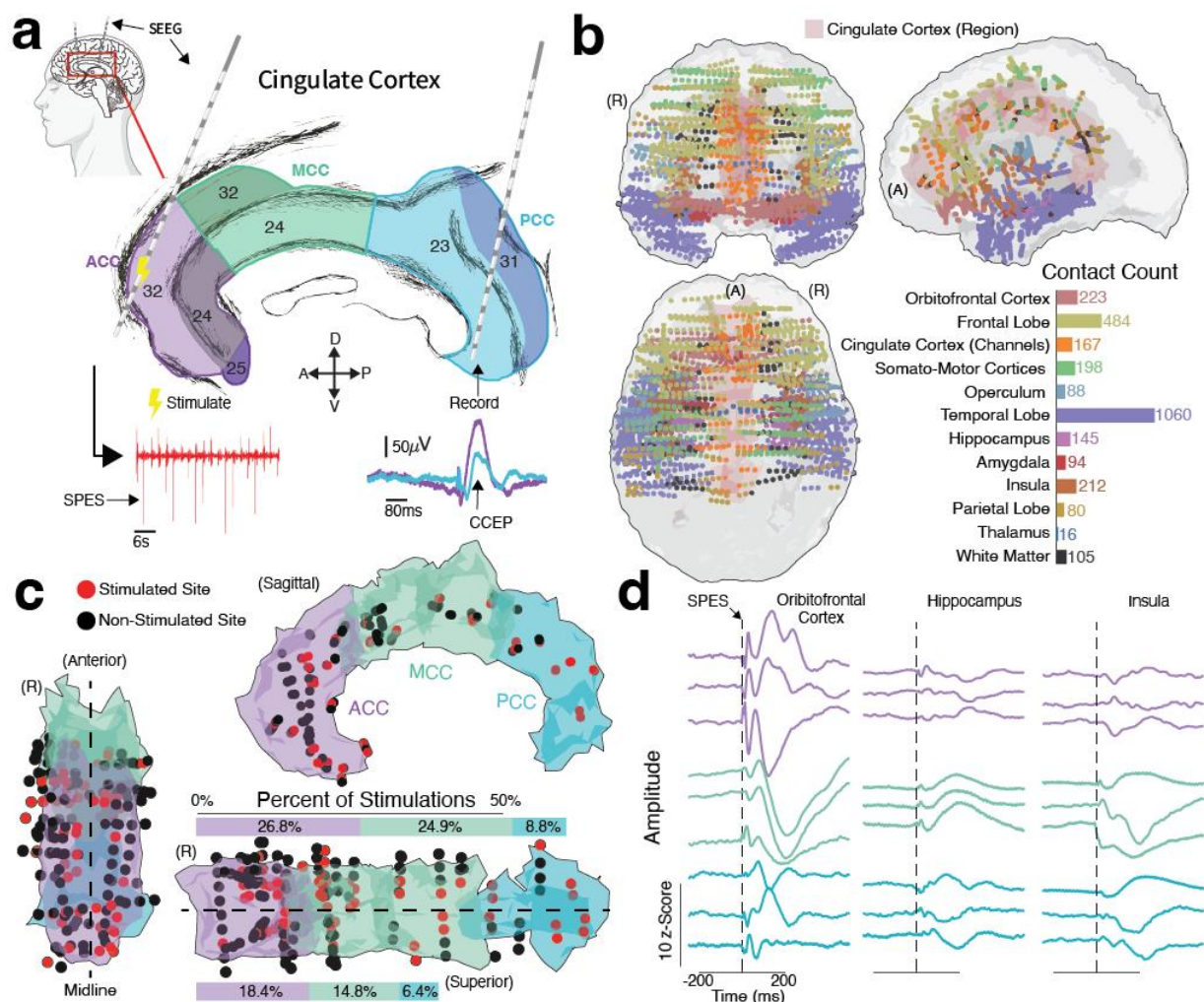


Figure 1. Experimental overview and anatomical coverage of SEEG recordings and stimulation. (a) Schematic of the experimental design. Patients with intractable epilepsy undergoing invasive SEEG monitoring received single-pulse electrical stimulation (6 mA, 0.2-ms biphasic square pulse, 0.5 Hz, 1-2 minutes) to the cingulate cortex (Cg) subregion(s). Cortico-cortical evoked potentials (CCEPs) were recorded from contacts distributed across various brain regions. (b) Anatomical coverage of SEEG contacts pooled from 15 patients, projected onto an MNI-space cortical model generated from volumetric averaging across all patients. The area of the Cg is shown in light red. Colors of contacts represent the major brain regions where contacts were located. (c) Position of contacts within the subregions of the Cg across patients, projected onto a volumetrically averaged MNI-space model, highlighting stimulated (red) and non-stimulated (black) contacts. Bar plots adjacent to the superior cortical view illustrate the distribution of stimulations across Cg subregions and hemispheres. (d) Exemplar CCEP waveforms across representative cortical regions, reflecting distinct morphological qualities elicited by stimulation of different Cg subregions.

CCEP Coherence as a Measure of Effective Connectivity

While CCEP magnitude ($CCEP_{mag}$) is useful for detecting regions effectively connected to the site of stimulation, it does not reveal CCEP morphological information, which is necessary for insight into the dynamics and functional properties of inter-regional connectivity (29). To this end, we implemented a CCEP coherence ($CCEP_{coh}$) measure, in addition to $CCEP_{mag}$ (measured as response RMS), to quantify the effective connectivity of the Cg subregions to other cortical areas. Since CCEPs resulting from SPES are characterized by detectable statistical changes in the local-field potential (LFP) morphology of regions effectively connected to stimulated tissue, the metric of $CCEP_{coh}$ complements measures of $CCEP_{mag}$ by providing measures of morphological consistency and diversity. Figure 2a demonstrates how $CCEP_{coh}$ is calculated. For each contact, a

set of unique pairwise correlations between stimulation events was calculated for the pre-stimulation window (baseline window) and the post-stimulation window (CCEP window). A Cohen's d between these two distributions represents the overall morphological consistency of the CCEPs recorded at each contact, relative to a distribution of baseline LFP (Figure 2b). We defined any contacts with a significant difference between these distributions as effectively connected to the stimulated region (Wilcoxon rank sum, $\alpha = 0.0001$, Benjamini-Hochberg FDR < 0.0001). Cortical surfaces showing the percentage of significant contacts within each brain region are shown in Supplemental Figure 1. Overall, this approach is used to represent statistical changes in the behavior of the LFP at regions effectively connected to each Cg subregion, where a greater Cohen's d represents a consistently evoked CCEP morphology, and a smaller Cohen's d represents a more diverse set of waveforms. A set of exemplar CCEP_{coh} distributions indicative of consistent morphology obtained from recordings in the orbitofrontal cortex is shown in Figure 2c.

We next determined the relationship between CCEP_{coh} (Cohen's d) and CCEP_{mag} (RMS). Supplemental Figure 2a shows a highly correlated relationship between these two metrics (Spearman's $\rho = 0.73$, $p < 0.0001$). Using receiver operating characteristic analysis, we evaluated which metric had greater performance in classifying significantly effectively connected contacts. Both metrics demonstrated large areas under the curve (0.99 for Cohen's d , 0.95 for RMS), showing high reliability in differentiating between significant and non-significant for both measures. For this study, we primarily report effective connectivity as CCEP_{coh} due to its greater area under the curve, its high correlation with CCEP_{mag}, and the utility of this measure for interpreting the morphological attributes of CCEPs across regions.

Effective Connectivity Networks of Cg Subregions. First, we evaluated the global effective connectivity of the Cg subregions by comparing the CCEP_{coh} across all significant contacts within each stimulation condition (n = 1030 for ACC, n = 1430 for MCC, n = 566 for PCC). Figure 2d

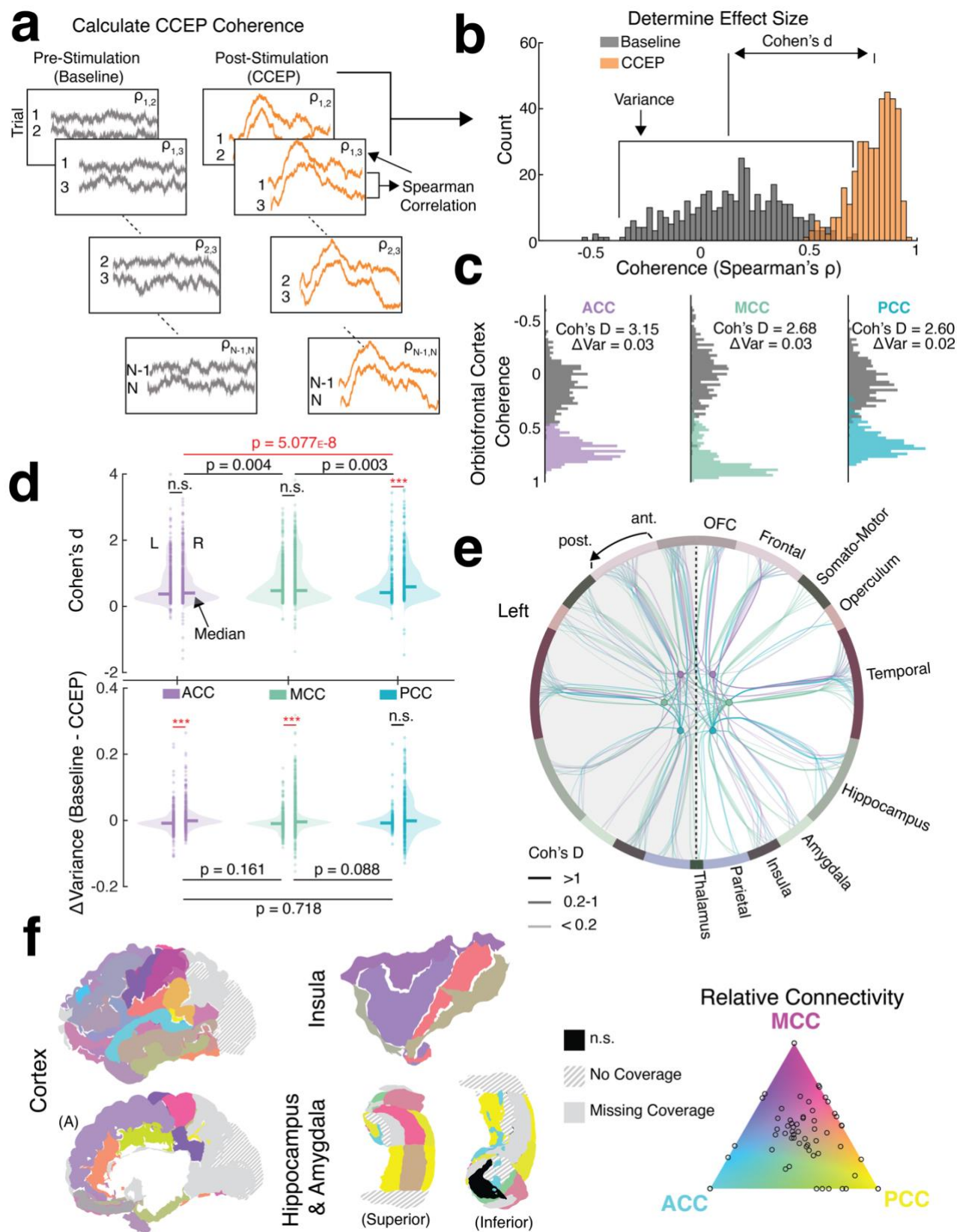


Figure 2. *Cingulate cortex effective connectivity modeled using cortico-cortical evoked potential (CCEP) coherence. (a & b)* Methodological overview of CCEP coherence calculation. **(a)** The inter-trial pairwise Spearman's correlations were calculated for the baseline and task CCEP windows, resulting in **(b)** the baseline and CCEP distributions of local field potential coherence for each contact. The effect size of connectivity between the stimulated cingulate cortex (Cg) subregion and the contact was defined as the Cohen's d between these distributions, representing the degree of separation between the single-pulse electrical stimulation (SPES) induced changes in the local field potential (LFP) morphology and the expected morphological similarity of pre-SPES LFP behavior. The variance of the CCEP distribution was subtracted from the variance of the baseline distribution to represent the variation in the inter-trial CCEP morphology corrected by the expected variation of inter-trial LFP morphology before SPES. **(c)** Exemplar baseline and CCEP coherence distributions recorded from representative contacts located in the orbitofrontal cortex after stimulation of each Cg subregion. **(d)** Global statistical comparison of Cohen's d (top) and Δ Variance (bottom) across all cortical contacts for each Cg stimulation condition: (L) or right (R) anterior, mid, or posterior cingulate (ACC, MCC, or PCC, respectively). Statistical differences were assessed using Wilcoxon rank sum tests ($\alpha = 0.001$). **(e)** Visualization of the divergent Cg effective connectivity network. Each circle in the middle represents a stimulation condition: either the left or right ACC (purple), MCC (green), or PCC (blue). The outer circle contains nodes representing each cortical region grouped into major anatomical areas, distinguished by unique colors. Cortical regions within each major anatomical area are ordered from anterior to posterior. Connectivity strength from stimulated Cg subregions to these cortical regions is represented by lines, whose thickness and opacity indicate the median Cohen's d across contacts within each region. **(f)** Cortical surface models visualize the relative effective connectivity of each Cg subregion to each cortical region. For each stimulation condition, the mean connectivity across all significant contacts within each region was calculated. The geometric mean of these three values was then plotted on a triangle, where the vertices of the triangle represent full connectivity to a Cg subregion. A color was assigned to each region based on an interpolated colormap across this triangle. Regions with similar connectivity to all three Cg subregions appear at the center of the triangle (greyish color). Regions with biased or preferential connectivity to one or more Cg subregions will appear closer to the triangle's edges, towards the vertices representing the subregions with greater connectivity.

shows a comparison of $CCEP_{coh}$ (mean $\pm$ SD, 0.61 ± 0.58 for ACC, 0.69 ± 0.61 for MCC, 0.74 ± 0.64 for PCC) and variance of the $CCEP_{coh}$ distributions relative to baseline (mean $\pm$ SD, 0.0012 ± 0.033 for ACC, 0.0026 ± 0.042 for MCC, 0.0062 ± 0.051 for PCC) across stimulation conditions. We observed a significant difference in $CCEP_{coh}$ between ACC and PCC stimulation conditions (Wilcoxon rank sum, $\alpha = 0.001$, $p = 0.004$ for ACC vs MCC, $p = 5.077E-8$ for ACC vs PCC, $p = 0.003$ for MCC vs PCC), which may indicate a proximity effect of CCEP morphological profiles (i.e. ACC having a statistically different $CCEP_{coh}$ profile from the PCC, but neither the ACC or PCC having a statistically different profile from the MCC). Further, there was a hemispheric difference in the $CCEP_{coh}$ distributions (mean $\pm$ SD, 0.61 ± 0.57 for left ACC, 0.61 ± 0.59 for right ACC, 0.62 ± 0.52 for left MCC, 0.73 ± 0.67 for right MCC, 0.63 ± 0.63 for left PCC, 0.79 ± 0.64 for right PCC) when comparing CCEPs resulting from stimulation of the left PCC and right PCC (Wilcoxon rank sum, $\alpha = 0.001$, $p = 0.35$ for ACC, $p = 0.13$ for MCC, $p = 1.14E-5$ for PCC). There were no statistically significant differences in relative variance between $CCEP_{coh}$ distributions (mean $\pm$ SD, -0.005 ± 0.03 for left ACC, 0.006 ± 0.03 for right ACC, -0.007 ± 0.03 for left MCC, 0.009 ± 0.05 for right MCC, -0.002 ± 0.03 for left PCC, -0.002 ± 0.03 for right PCC) within the three stimulation conditions; however, there were hemispheric differences in the ACC and MCC stimulation conditions (Wilcoxon rank sum, $\alpha = 0.001$, $p = 1.47E-13$ for ACC, $p = 8.45E-15$ for MCC, $p = 0.02$ for PCC).

Figure 2e shows the entirety of the divergent Cg effective connectivity network. This visualizes the effective connectivity of each Cg subregion to each cortical region available from our study data. Additionally, we evaluated the inter-Cg subregion effective connectivity (Supplemental Figure 2b) to determine whether there was a proximity effect of connectivity

between subregions. We found no significant differences in the $CCEP_{coh}$ between the ACC to the MCC and the PCC, nor the PCC to the MCC and the ACC. Instead, there was a significant difference in connectivity between the MCC to the ACC and PCC (Wilcoxon rank sum, $\alpha = 0.001$, $p = 7.84E-6$, mean $\pm 95\%$ CI, 0.40 ± 0.10 MCC to ACC, 1.56 ± 0.38 MCC to PCC).

Relative Connectivity of Cortical Regions to Each Cg Subregion

To determine the connectivity profile of each cortical region to each of the Cg subregions, we implemented a relative connectivity analysis. The results of this analysis are visualized in Figure 2f. For each region, we calculated a geometric mean effective connectivity value for each of the Cg subregions. This enabled the identification of cortical areas with preferential and non-preferential effective connectivity to each Cg subregion. We found a highly heterogeneous relative connectivity profile across cortical regions. Notably, frontal regions showed higher connectivity to the ACC and MCC, and parietal regions showed higher connectivity to the MCC and PCC. The insula cortex, specifically the short insular gyrus and the superior insula, showed dominant effective connectivity to the ACC and MCC. The long insular gyrus had substantially higher effective connectivity to the MCC and PCC than to the ACC. There were also several regions with near-complete effective connectivity dominance to single Cg subregions. Examples include subregions of the hippocampus, which were predominantly connected to the PCC, the motor cortices, which were only effectively connected to the MCC, and the superior temporal gyrus, which was only effectively connected to the ACC.

This analysis enabled us to identify regions that had highly similar effective connectivity to each of the Cg subregions (within 5% of the centroid of the triangular connectivity plane; for details, see “Relative Cingulate Cortex Connectivity” in Methods). When using $CCEP_{coh}$ to model effective connectivity, these regions included the middle-frontal gyrus, the triangular part of the inferior frontal gyrus, gyrus rectus, the anterior segment of the circular sulcus of the insula, the middle temporal gyrus, and the temporal pole. Extended cortical surface models showing the effective connectivity of each region to each Cg subregion using CCEP are shown in Supplemental Figure 2c.

The relative Cg subregion effective connectivity modeled using $CCEP_{mag}$ is shown in Supplemental Figure 3a. While the relative connectivity profile across cortical regions was similar to what was found using $CCEP_{coh}$, there were notable differences. Namely, several regions of the hippocampus showing significant $CCEP_{coh}$ were not significant when using $CCEP_{mag}$ as the metric for effective connectivity across stimulation conditions (Supplemental Figure 3b). We found that the middle-frontal gyrus, the triangular part of the inferior frontal gyrus, gyrus rectus,

inferior segment of the circular sulcus of the insula, and lateral nucleus of the amygdala had highly similar effective connectivity to each Cg subregion.

Together, these results provide a comprehensive cortical map of Cg effective connectivity, using both CCEP_{coh} and CCEP_{mag}. Further, they show the complementary utility of these two metrics, as while some regions may not demonstrate significant changes in CCEP_{mag}, significant changes in LFP morphology can still be detected using CCEP_{coh}.

Spatial Distributions of Effective Connectivity Across Cg Subregion Effective Connectivity Networks

We next characterized the spatial profile of Cg subregion connectivity across the cortex. We found that across all three spatial axes (anterior-posterior, superior-inferior, and left-right), the distributions of significantly effectively connected contacts in each stimulation condition showed significant spatial clustering (Figure 3a, Moran's I statistic, 1,000 permutation significance test, $\alpha = 0.01$, $p < 0.001$ for all conditions). We next evaluated whether the spatial clustering, specifically the centroid of these spatial distributions, was significantly different between stimulation conditions across each spatial axis. We found that there was only a significant difference in centroid positions along the anterior-posterior axis, specifically between the ACC and MCC centroids (permutation-based pairwise comparison of centroid differences, $\alpha = 0.05$, $p = 0.008$) and ACC and PCC ($p = 0.002$) centroids. The expanded spatial distributions and permutation distributions are shown in Supplemental Figure 4. These results reflect a clear anterior-posterior effective connectivity gradient, with the centroid of the ACC distribution being most anterior and the PCC distribution centroid being the most posterior.

We next aimed to characterize the hemispheric differences in effective connectivity across major brain region groups, contralateral and ipsilateral to the stimulated Cg subregion (Figure 3b). For regions contralateral to the stimulated subregion, we found significantly higher effective connectivity from the ACC to the right orbitofrontal cortex (Wilcoxon rank sum, $\alpha = 0.05$, $p < 0.05$) and from the MCC to the left frontal cortex ($p < 0.001$). There were insufficient contralateral contacts in the operculum, temporal lobe, hippocampus, amygdala, parietal lobe, and thalamus for statistical comparisons due to limitations in the clinical coverage of this patient population. We found no significant differences between hemispheres for the somato-motor cortices or the insular cortices in this contralateral comparison. For comparison of major region groups ipsilateral to the stimulated Cg subregion, we found significantly higher effective connectivity (i.e., hemispheric dominance) from the MCC to the right orbitofrontal cortex ($p < 0.001$), significantly higher effective connectivity from the ACC and MCC to the left frontal

cortex (ACC $p < 0.05$, MCC $p < 0.001$), significantly higher effective connectivity from the PCC to the right operculum ($p < 0.05$), significantly higher effective connectivity from the ACC, MCC, and PCC to the right temporal lobe (ACC $p < 0.001$, MCC $p < 0.001$, PCC $p < 0.05$), significantly higher effective connectivity from the PCC to the right hippocampus ($p < 0.05$), and significantly higher connectivity from the MCC and PCC to the right insula (MCC $p < 0.05$, PCC $p < 0.001$). A cortical surface visualization showing the average hemispheric difference of contacts within each region across stimulation conditions is shown in Supplemental Figure 5. Notably, only frontal regions demonstrated a left hemisphere dominance in effective connectivity for ACC and MCC conditions, with most other regions having right hemisphere dominance across stimulation conditions.

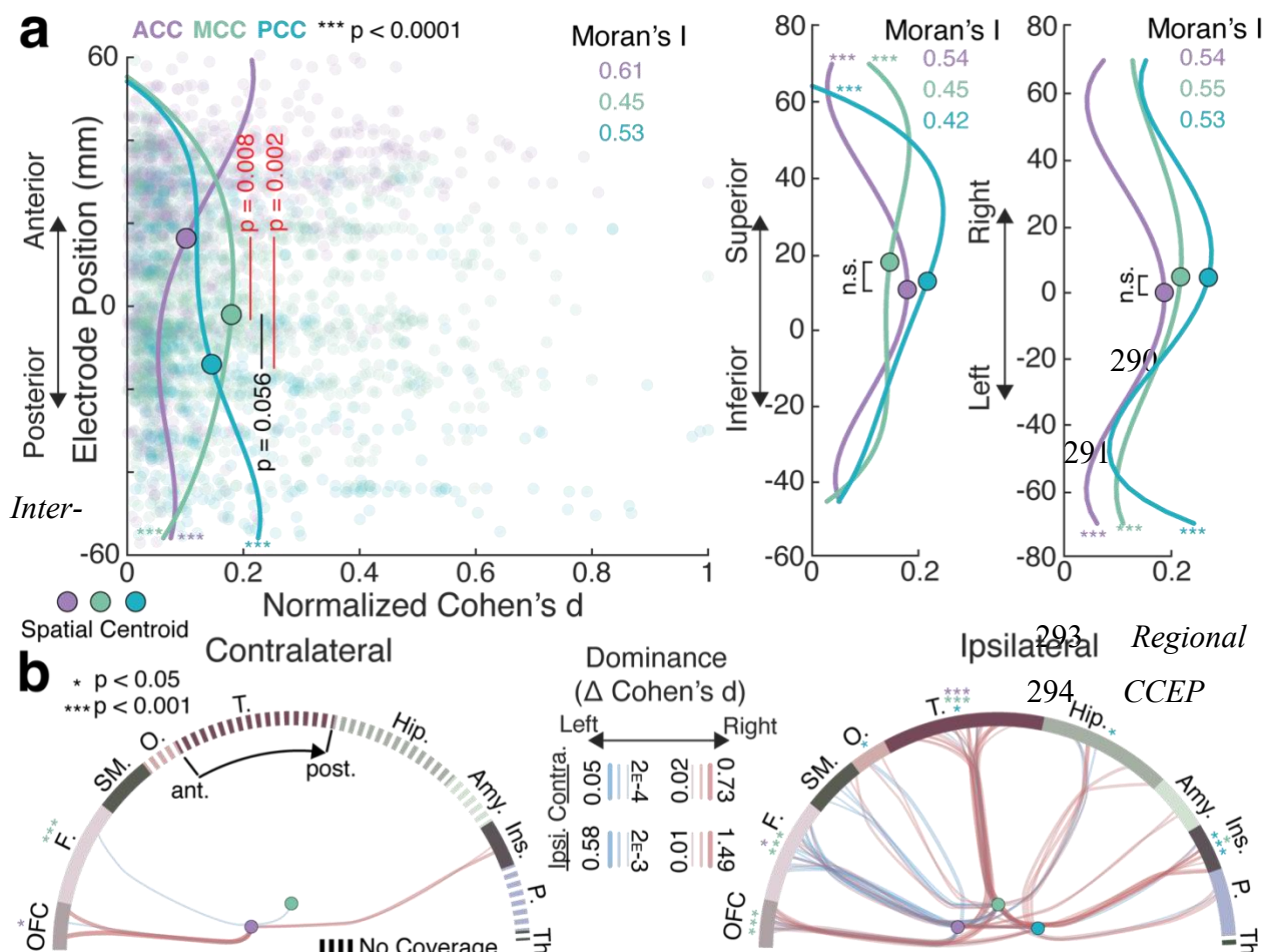


Figure 3. Spatial Distribution of Effective Connectivity Across the Cortex. **(a)** Distributions of normalized effective connectivity (Cohen's d) across the anterior-posterior, inferior-superior, and left-right axes. Lines show a 4th-order polynomial trend line fitted to the mean Cohen's d of 50 evenly spaced bins for each spatial dimension. The circles along the lines represent the centroids of each effective connectivity distribution. Moran's I was calculated to quantify biases in connectivity across each spatial dimension for each condition. Permutation-based pairwise comparisons of observed centroid differences were calculated to compare centroids of distributions for each stimulation condition ($\alpha = 0.05$). **(b)** Network wiring diagrams showing hemispheric differences in connectivity for regions contralateral (left) or ipsilateral (right) to the site of stimulation for each condition. Blue lines represent greater connectivity in the left hemisphere, and red lines represent greater connectivity in the right hemisphere. The thickness of lines represents the magnitude of the difference between the average connectivity strength of significant contacts within the right and left regions. Statistical tests between left and right hemisphere regions were conducted on contacts within major region groups: orbitofrontal cortex (OFC), frontal cortex (F), somato-motor cortices (SM), operculum (O), temporal lobe (T), hippocampus (Hip), amygdala (amy), insula (Ins), parietal lobe (P), thalamus (Th) ($\alpha = 0.05$, * $p < 0.05$, *** $p < 0.001$). Dashed lines on the outer network semicircle represent major region groups without substantial contacts for statistical comparisons.

Morphology-based Hierarchical Clustering Models of Cg Subregion Effective Connectivity Networks

Beyond a divergent model of Cg effective connectivity and characterizing CCEP morphological consistency across regions using $CCEP_{coh}$, we also aimed to evaluate the morphological similarity of CCEPs among cortical regions. To this end, we implemented an inter-region cross-correlation analysis to evaluate the morphological similarity of CCEPs between cortical regions for each Cg subregion's effective connectivity network. We then used hierarchical clustering to group regions by the morphological similarity of CCEPs for each stimulation condition (Supplemental Figure 6). The results of this analysis are shown in Figure 4, revealing three highly differentiable inter-regional correlation matrices across stimulation conditions. Our analysis revealed eight clusters for the ACC (Figure 4a_i and a_{ii}), six clusters for the MCC (Figure 4b_i and b_{ii}), and seven clusters for the PCC (Figure 4c_i and c_{ii}). A full list of regions within each cluster for each stimulation condition is described in Supplemental Results. Notably, the MCC contained the largest single cluster, accounting for 30 of the 45 regions included in the hierarchical clusters. It also appeared that there was no major spatial organization of clusters across stimulation conditions, indicating that the morphological signature of CCEPs is not clustered in space.

We next compared the average pairwise morphological similarity of regions within each correlation matrix between stimulation conditions. The MCC network had the highest mean pairwise correlation values across regions (mean $\pm 95\%$ CI, 0.37 ± 0.04) and was significantly different from both the ACC network (ACC 0.32 ± 0.04 , Wilcoxon rank sum, $\alpha = 0.05$, $p = 0.018$) and PCC network (PCC 0.29 ± 0.03 , $p = 2.42E-4$). There was no significant difference between the ACC network and the PCC network ($p = 0.129$).

We next evaluated the diversity of inter-regional CCEP morphology across Cg subregion effective connectivity networks. To do this, we performed a comparison of the cophenetic distance distributions of each network's dendrogram. The PCC network dendrogram contained the highest average cophenetic distance (mean $\pm SD$, 0.69 ± 0.20) and lowest cophenetic entropy (0.326). The ACC and MCC network dendrograms were highly similar (ACC cophenetic distance $0.57 \pm .22$, cophenetic entropy 0.390; MCC cophenetic distance 0.54 ± 0.24 , cophenetic entropy 0.395). Pairwise comparisons of these distributions revealed a significant difference between all three stimulation conditions (Wilcoxon rank sum, $\alpha = 0.05$, $p < 0.001$ for comparisons).

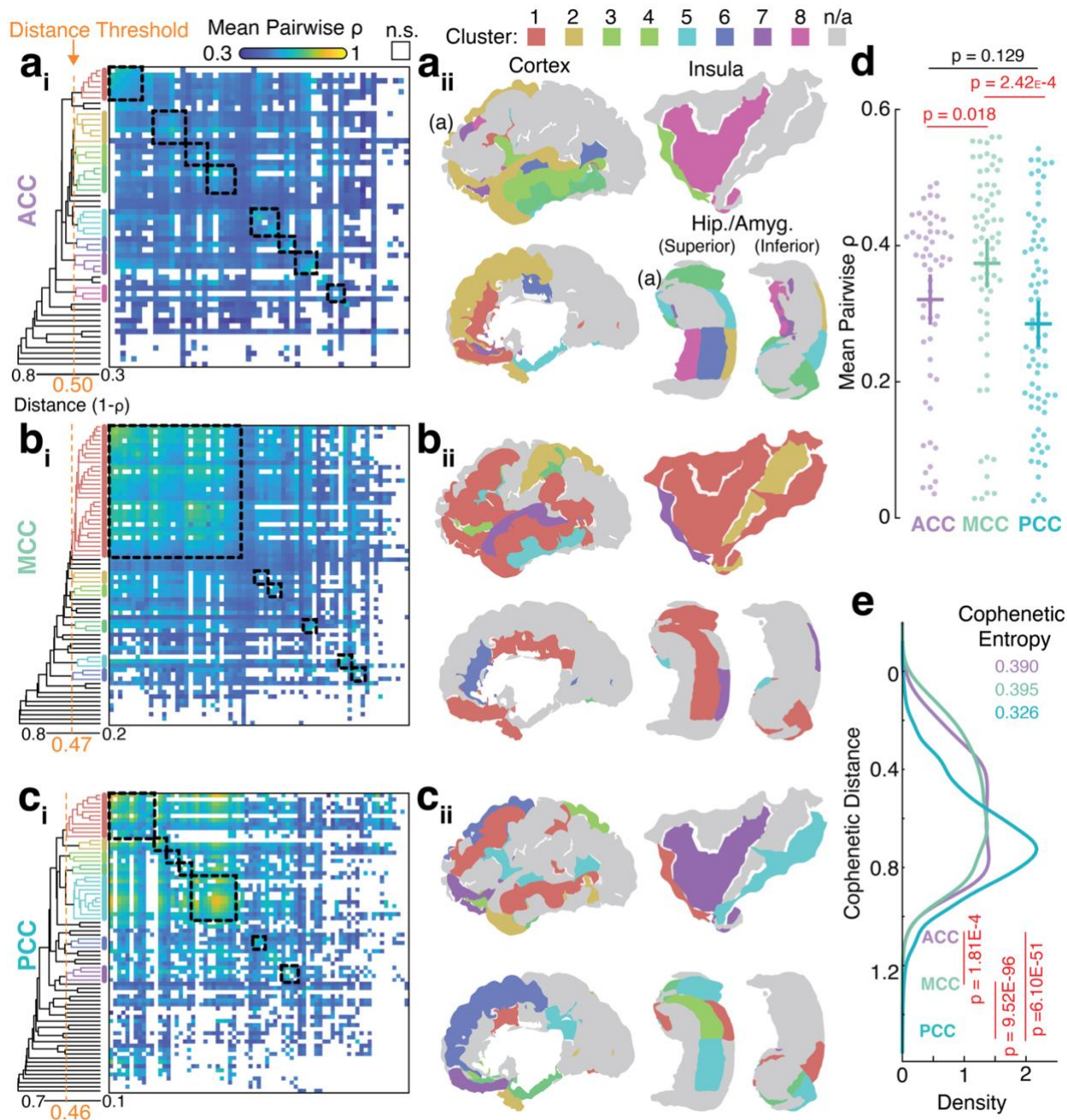
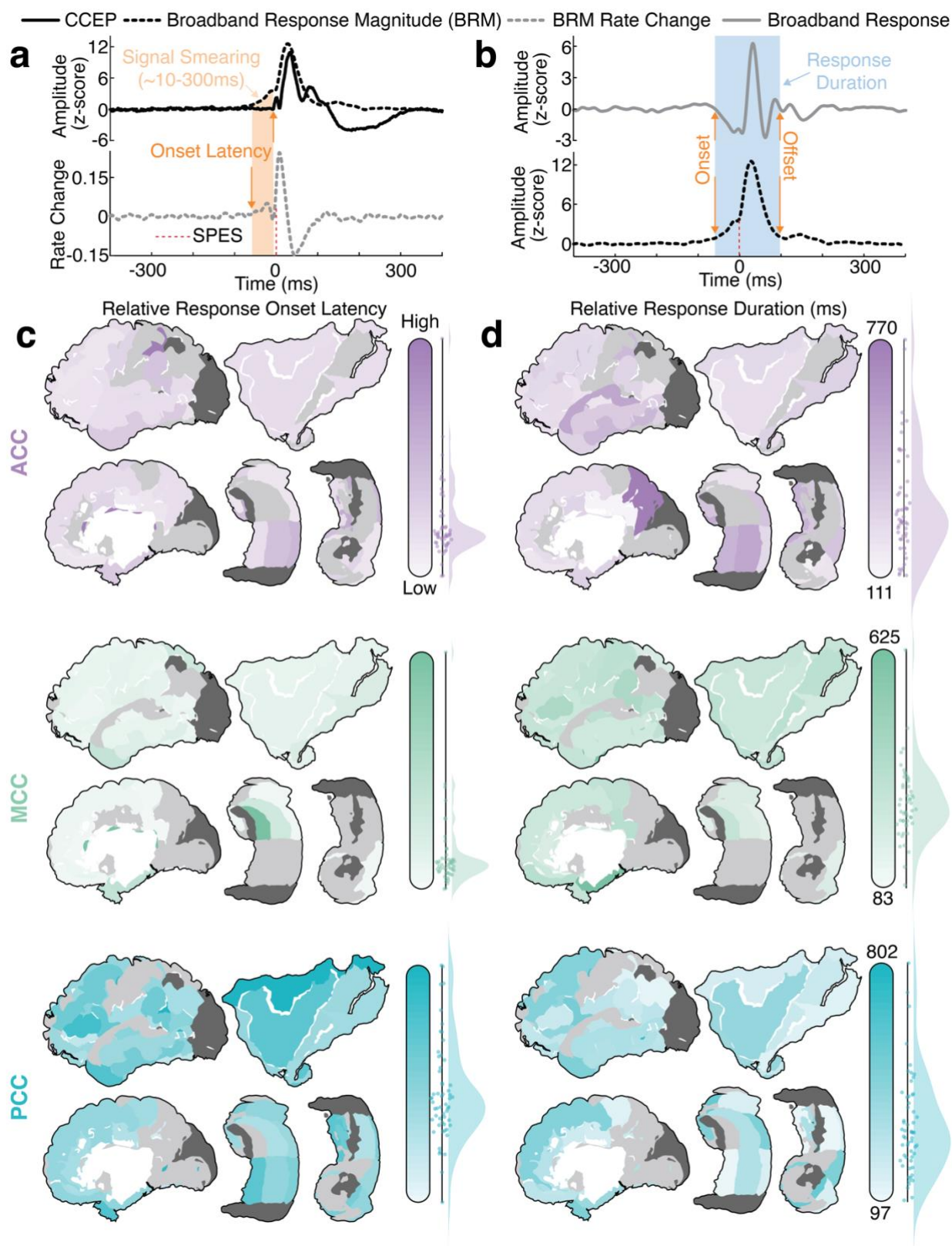


Figure 4. *Hierarchical Clustering of Cingulate Cortex Effective Connectivity Networks.* (a-c) Cingulate cortex (Cg) subnetworks are defined and organized through hierarchical clustering of cortico-cortical evoked potential (CCEP) morphology. (a) Subnetworks of the anterior cingulate cortex (ACC). (a_i) Correlation matrix showing the mean inter-regional similarity of CCEP morphology, organized from highest to lowest similarity. Dotted boxes show clusters. White entries in the correlation matrix represent regions without statistically significant CCEPs (Wilcoxon rank sum test, $\alpha = 0.0001$). The adjacent dendrogram shows the hierarchical structure of the correlation matrix. Clusters of regions on the dendrogram are colored by cluster number, from highest to lowest intra-cluster similarity. The optimal distance threshold to resolve clusters is shown by the dashed orange line. (a_{ii}) Anatomical location of clusters. (b) Mid-cingulate cortex (MCC) subnetworks. (c) Posterior cingulate cortex (PCC) subnetworks. (d) Comparison of inter-regional CCEP morphological similarity across stimulation conditions (Wilcoxon rank sum, $\alpha = 0.05$). (e) Comparison of dendrogram tree diversity as a distribution of cophenetic distance across stimulation conditions (Wilcoxon rank sum, $\alpha = 0.05$).

Temporal Dynamics of the Cg Subregion Effective Connectivity

We implemented a changepoint detection algorithm to determine the average onset/offset latency and duration of CCEPs across regions for each stimulation condition (Figure 5a & 5b). Figure 5c shows the normalized response latency across brain regions in each stimulation condition. CCEPs

330 in the ACC and MCC conditions showed homogeneity in onset time, with higher latency
331 responses occurring in the hippocampus and at the temporal poles. The ACC also showed higher
332 latency responses in the postcentral sulcus, the inferior temporal lobe, and the supramarginal
333 gyrus. The PCC condition showed a wider spread of response onset latencies, with later responses
334 occurring in the central and superior insula, frontal lobe, inferior and posterior temporal lobe, and
335 parietal lobe. We found a significant positive correlation between response latency and distance
336 from the recording electrode to the stimulating electrodes (Supplemental Figure 7; Spearman's
337 $\rho = 0.33$, $p < 0.0001$). Response duration showed a similar comparative trend between regions,
338 with the most variable spread of high-duration CCEPs in the PCC stimulation condition (Figure
339 5d). The ACC condition had the longest responses in the hippocampus, postcentral sulcus,
340 superior temporal lobe, and precuneus. The MCC condition had the longest responses in the
341 parahippocampal gyrus, inferior frontal gyrus, and supramarginal gyrus. The PCC condition
342 showed high duration responses throughout the frontal lobe, hippocampus, temporal pole, and
343 short insular gyri.



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Figure 5. *Temporal Dynamics of Cingulate Cortex Effective Connectivity.* **(a)** Cortico-cortical evoked potential (CCEP) onset was detected using a time-series optimal partitioning algorithm (see Methods) to identify the first instance relative to stimulation, where the statistical properties of the broadband (5–40 Hz) local field potential (LFP) slope changes from baseline. For this analysis, the temporal distortion caused by bandpass filtering created a spectral smearing artifact, resulting in the detected onset of the CCEP to occur prior to stimulation in many cases. To account for this, the onset was normalized across conditions, enabling comparison of response onset latencies between regions, at the cost of accurate onset values. **(b)** Response duration was calculated using both the response onset and the response offset. The response offset was calculated using the optimal partitioning method on the broadband CCEP response magnitude, specifically, by identifying the two indices that best separate the statistical properties of the broadband CCEP. The second of these two indices was used as the CCEP offset. **(c)** Anatomical visualization of normalized response onset latencies across regions for each stimulation condition. Darker colors indicate later CCEP onsets. Distributions of onset latencies across regions are shown adjacent to the colorbar. **(d)** Anatomical visualization of response duration across regions for each stimulation condition, where darker colors represent longer responses.

351 *Stimulation-Evoked Gamma Responses across Cg Subregions*

Beyond modeling Cg effective connectivity through SPES-induced CCEPs, we also evaluated evoked broadband gamma to characterize average changes in local neuronal firing within each region. Figure 6a shows the average evoked broadband gamma profiles for each stimulation condition calculated from contacts showing both significant effective connectivity and significant changes in broadband gamma. Most regions showed a SPES-induced increase in broadband gamma in the ACC and MCC conditions. The greatest increase in broadband gamma power was found in the hippocampus for the ACC condition, and the middle frontal gyrus in the MCC condition. The PCC evoked broadband gamma response profile was heterogeneous, showing increases and decreases in broadband gamma power across the cortex. Notable increases in broadband gamma occurred at the temporal pole, ACC, posterior MCC, and the superior segment of the circular sulcus of the insula. Notable decreases were observed in the superior frontal gyrus, the hippocampal body, the short insular gyri, and the anterior segment of the circular sulcus of the insula. Figure 6b shows the proportion of contacts determined to be significantly effectively connected to the Cg subregions that also showed a significant SPES-induced change in broadband gamma (64.3% for ACC, 66.8% for MCC, 55.7% for PCC).

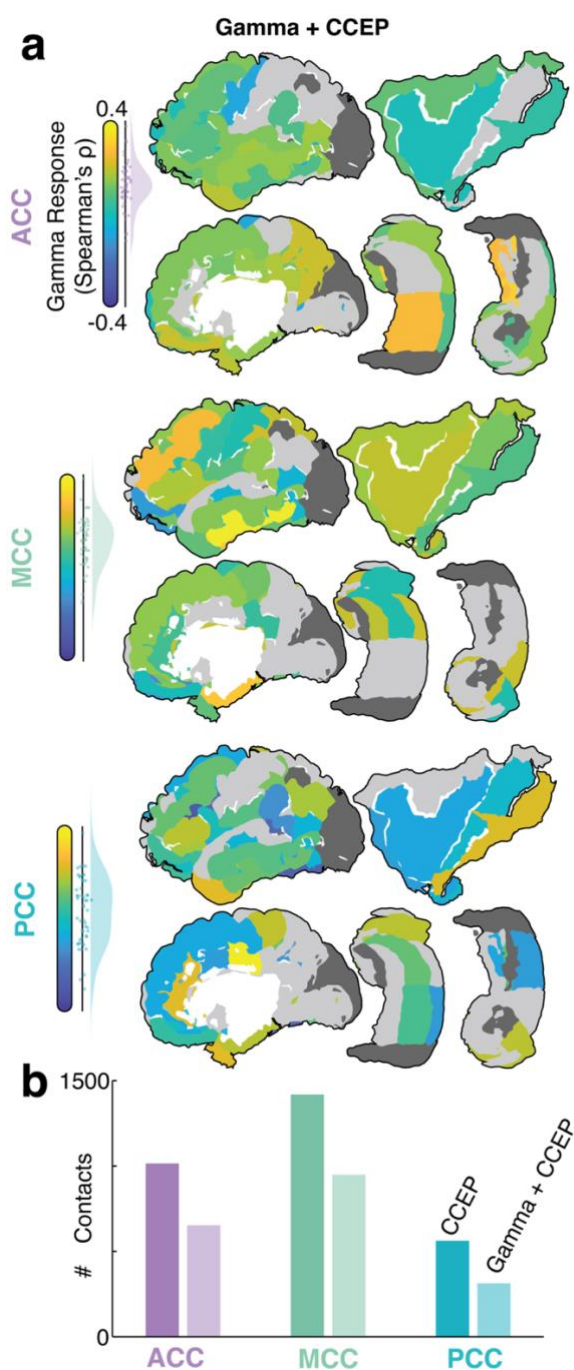


Figure 6. *Gamma Response Profiles of the Cingulate Cortex Effective Connectivity Network.* (a) Anatomical profile of average single-pulse electrical stimulation evoked broadband gamma responses concurrent with significant cortico-cortical evoked potentials (CCEPs). Increases in gamma are shown in yellow, decreases are shown in blue. (b) Proportions of contacts showing both significant effective connectivity to each cingulate cortex subregion and an evoked broadband gamma response.

Supplemental Figure 8 compares the evoked gamma responses observed across all conditions and between hemispheres. There were significant differences between all three conditions ($p = 0.003$ for ACC vs MCC, $p = 7.12\text{E-}9$ for MCC vs PCC, $p = 6.33\text{E-}18$ for ACC vs PCC). There were also significant hemispheric differences observed between left and right Cg stimulation for the MCC ($p = 4.86\text{E-}29$) and the PCC ($p = 0.006$). Supplemental Table 2 lists the statistical information comparing the ipsilateral evoked gamma in response to stimulation of left or right Cg subregions. Notably, there was a significant difference in evoked gamma between the left and right frontal lobes for the ACC condition ($p = 0.005$), frontal lobes, insula, orbitofrontal cortex, and temporal lobes for the MCC condition ($p = 8.38\text{E-}7$, $p = 5.61\text{E-}5$, $p = 5.59\text{E-}8$, $p = 0.004$, respectively), and the frontal lobes for the PCC condition ($p = 3.39\text{E-}5$). Several regions also demonstrated a difference in gamma response direction (increased or decreased gamma) between hemispheres (Supplemental Table 3). For the MCC condition, gamma responses increased in the left orbitofrontal cortex ($\text{Rho} = 0.22$) and decreased in the right ($\text{Rho} = -0.07$). For the PCC condition, there was an increase in the left frontal lobe (0.06) and a decrease in the right (-0.28).

400 Finally, we evaluated the importance of features used to profile the Cg effective connectivity
401 networks. To do this, we determined the predictive strength of features in classifying the
402 stimulated Cg subregion in a random forest model. Features were characterized into 6 main
403 groups (for details, see Assessing Feature Importance for Classifying Cg Effective Connectivity
404 Networks in Methods): effective connectivity, spatial, waveform (assumption-free), evoked
405 gamma, temporal, and waveform (assumption-based). Figure 7a shows the confusion matrix
406 outlining the performance of our model in predicting which Cg subregion was stimulated at each
407 contact. The overall accuracy of the model was 75.5%. The classification accuracy was 78.9% for
408 ACC contacts, 73.1% for MCC contacts, and 72% for PCC contacts. Figure 7b shows the
409 predictor strength of features ranked from highest to lowest. CCEP_{mag} (RMS) was the highest
410 predictor, followed by each of the three spatial coordinates of the contacts. Measures of effective
411 connectivity were all ranked in the top 50% of features for predictor strength. Notably,
412 assumption-based waveform features, which are commonly used to characterize effective
413 connectivity, had the lowest predictor strength. Figure 7c shows the average contact classification
414 accuracy across cortical regions. Most regions had high classification accuracy. The only regions
415 with chance or less than chance accuracy were in the hippocampus and amygdala. Correctly
416 classified contacts showed significantly higher RMS and Cohen's d than incorrectly classified
417 (Figure 7d; mean $\pm$ SD, correct RMS 0.49 ± 0.72 , incorrect RMS 0.32 ± 0.31 , Wilcoxon rank sum,
418 $\alpha = 0.05$, $p < 0.0001$; correct Cohen's d 0.21 ± 0.45 , incorrect Cohen's d 0.11 ± 0.27 , $p < 0.0001$).
419 Correctly classified contacts were also shown proportionately greater significance than incorrectly
420 classified contacts for both RMS and Cohen's d (Figure 7e; correct RMS 18.0%, incorrect RMS
421 9.27%; correct Cohen's d 26.0% incorrect Cohen's d 18.5%). While this trend demonstrates that
422 statistical significance may play a role in representing effective connectivity, the overall
423 classification accuracy across contacts was substantially higher than the proportion of significant
424 contacts across conditions (15.8% of contacts significant for RMS, 24.4% of contacts significant
425 for Cohen's d).

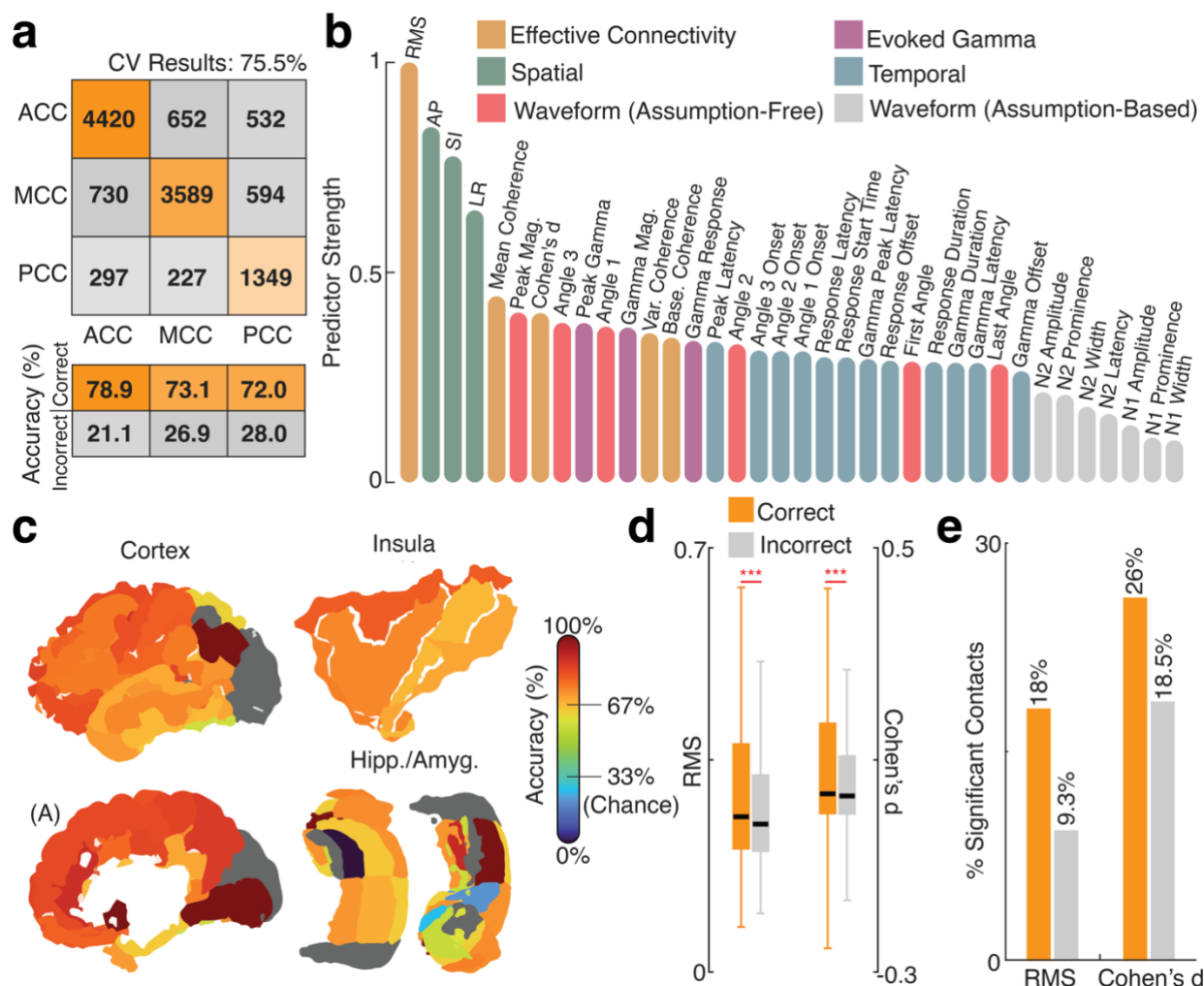


Figure 7. Assessing Feature Importance for Classifying Effective Connectivity to Cingulate Cortex Subregions. (a) Confusion matrix evaluating the classification performance of a Random Forest Model trained to classify contacts by which cingulate cortex subregion was stimulated. (b) Predictive strength of input features in classifying which cingulate subregion was stimulated, indicating the impact of each feature on the classification model after random permutation. Features were divided into six feature classes: effective connectivity features, spatial features, assumption-free waveform features, evoked gamma features, temporal features, and assumption-based waveform features. Features (in descending order of predictor strength) included CCEP amplitude as root-mean-squared (RMS), anterior-posterior contact position (AP), superior-inferior contact position (SI), left-right contact position (LR), mean CCEP coherence (Mean Coherence), peak broadband CCEP magnitude (Peak Mag.), CCEP coherence as Cohen's d (Cohen's d), third angle of the broadband CCEP waveform (Angle 3), peak broadband gamma response magnitude (Peak Gamma), first angle of the broadband CCEP morphology (Angle 1), average magnitude of the broadband gamma response (Gamma Mag.), difference between the variance of the baseline and CCEP coherence distributions (Var. Coherence), average coherence of the baseline coherence distribution (Base. Coherence), gamma response as Spearman's rho (Gamma Response), latency of the peak broadband CCEP magnitude (Peak Latency), second angle of the broadband CCEP waveform (Angle 2), latency of the third angle of the CCEP waveform (Angle 3 Onset), latency of the second angle of the CCEP waveform (Angle 2 Onset), latency of the first angle of the CCEP waveform (Angle 1 Onset), latency of the broadband CCEP waveform onset detected using the first derivative of the broadband magnitude (Response Latency), latency of the broadband CCEP waveform onset detected using the broadband magnitude (Response Start Time), latency of the peak evoked gamma response (Gamma Peak Latency), latency of the broadband CCEP waveform offset (Response Offset), angle of the CCEP waveform response onset (First Angle), duration of CCEP calculated from the magnitude of the broadband CCEP waveform (Response Duration), duration of evoked gamma response (Gamma Duration), latency of the evoked gamma response onset (Gamma Latency), angle of the CCEP waveform response offset (Last Angle), evoked gamma response offset (Gamma Offset), N2 peak amplitude (N2 Amplitude), N2 peak prominence (N2 Prominence), N2 peak width (N2 Width), N1 peak amplitude (N1 Amplitude), N1 peak prominence (N1 Prominence), N1 peak width (N1 Width). (c) Classification accuracy expressed as a percentage of contacts correctly classified for each brain region, across brain regions. (d) Comparison of primary effective connectivity measures, RMS, and Cohen's d between correctly and incorrectly classified contacts (Wilcoxon rank sum test, $\alpha = 0.05$, *** $p < 0.001$). (e) Differences in the percentage of significant contacts for correctly and incorrectly classified contacts across primary effective connectivity measures.

Discussion

Our study presents a comprehensive, multi-modal effective connectivity profile of the human Cg modeled using invasive SPES of 15 resection-free human patients with refractory epilepsy. The present study provides direct evidence of the differential topographical organization of the ACC, MCC, and PCC effective networks. We demonstrate a clear spatial organization of these networks, specifically along an anterior-posterior axis, which mirrors anatomical organization. We also revealed the differential hierarchical organization of effective networks, as defined by CCEP morphology. Further, by extracting and comparing several statistical features of CCEPs, we confirm that effective connectivity is measurement-dependent and that conventional methods, such as response magnitude, are not sufficient in fully characterizing effective networks or detecting connections to stimulated regions.

Key methodological strengths of the present study include constant stimulation parameters chosen to maximize response signal across recording contacts. Previous titration studies have demonstrated that CCEPs are dependent on stimulation parameters and that stimulation currents of at least 6 mA are necessary to maximize the magnitude and spatial distribution of responses (27, 33). Further, our study does not include any data from patients with major brain resections or a seizure onset zone in the Cg, thereby preserving native anatomy and white-matter pathways that underlie effective connections (34).

We quantified effective connectivity with two complementary metrics: CCEP magnitude, which captures the strength of connection (and is widely used in the literature), and CCEP coherence, which measures the consistency of the evoked waveform to stimulation. While CCEP magnitude is a suitable measure for identifying responses based on the strength of connection, CCEP coherence serves as a measure of response waveform morphological consistency or heterogeneity. This measure is an alternative to isolating the most common components of evoked waveforms, which is critical to understanding highly specific interactions between regions, especially in hypothesis-preselected CCEP studies supported by basis profiling and canonical response parameterization (29, 32). We instead leverage this metric of coherence to represent the diversity of observed waveform morphology at sites effectively connected to the stimulated target, which better serves comparisons of whole-network profiles. If effective connectivity can be defined as the influence that one neuronal population exerts over another (23), then the morphological changes in LFP following SPES provide insight into the function and physiological mechanisms related to how the stimulation site communicates with downstream regions. Thus, this measure represents the diversity of accessible signaling or communication between effectively connected regions.

By implementing CCEP coherence as a measure of effective connectivity, we were able to detect changes not only in the magnitude of the LFP, but also in how uniformly that response's morphology is reproduced across stimulation events. This measure's utility was evident when defining effective connections from the Cg subregions to the hippocampus. We found that several hippocampal subregions showed significant effective connectivity when using CCEP coherence, but not when using CCEP magnitude. This result may reflect highly variable hippocampal activity (e.g., sharp-wave ripples), which prevent detectable quantitative changes in LFP magnitude after stimulation, or unique stimulation-induced changes in LFP morphology due to the geometric relationship between recording electrodes and the hippocampus's curved anatomical structure (39, 40). This indicates that while the magnitude of the LFP after stimulation was not statistically different from pre-stimulation, changes in CCEP morphological characteristics were detectable through our CCEP coherence metric.

Our models of Cg effective connectivity showed widespread connections with prominent responses in areas coinciding with previous literature. The ACC showed predominant connections to orbito-frontal, frontal, mesial temporal, temporal, and insular structures, reproducing previous stimulation studies and supporting functional and anatomical literature (4, 41). Interestingly, of all three stimulation conditions, the ACC was the only subregion to show significant connections to the superior temporal gyrus, possibly reflecting the ACC's top-down attentional and conflict-monitoring modulation of auditory and language processing (42, 43). While functional and anatomical links between these two regions have been noted (4, 44, 45), to our knowledge, the physiology and role of this effective connection have yet to be studied. The MCC's predominant connections to frontal, somato-motor, insular, and mesial temporal structures, and the PCC's connections to frontal, parietal, insular, and mesial temporal structures also support previous findings (4, 41).

We were able to profile the relative connectivity of each cortical region to each of the Cg subregions. Through this analysis, we found that the middle-frontal gyrus, the triangular part of the inferior frontal gyrus, and the gyrus rectus exhibited near-equivalent effective connectivity with all three CC subregions. This was true for both of our effective connectivity measures, meaning that the observed CCEPs in these regions had near-equal magnitude (directional connectivity) and coherence (morphological diversity) (28). This finding reveals the existence of Cg "hubs," or regions with effective overlap between all three Cg subregions. Specifically, the highly similar nature of connective input from each cingulate subregion to these regions (in both magnitude and morphological diversity) suggests these regions may play a similar computational or functional role in all three Cg effective networks.

stereotyped behaviors (18). Interestingly, temporal analysis revealed that CCEPs in the MCC condition were low-latency and homogenous, which further supports this idea. Thus, it is possible that out of the three Cg subregions, the MCC network has the most stereotyped set of communication characteristics and thus requires less diversity in how it communicates with other regions when compared to the ACC or PCC. This, of course, is preliminary and speculative, and does not account for the battery of complex cognitive functions of the MCC, which likely require diverse communication protocols between regions. However, such studies evaluating the relationship between network function and diversity of inter-regional neural interactions can be supported through this analysis.

This inter-regional analysis also presents CCEP clusters organized by morphological similarity between regions. We found that the morphology of some clusters followed the N1/N2 shape often described in contemporary analyses of CCEPs (26, 28, 58, 59), especially clusters with the highest inter-regional morphological similarity. However, other clusters expressed CCEP morphologies beyond the commonly assumed N1/N2 shape. Others have noted CCEP morphologies that do not rely on pre-defined assumptions about shape, the limitations of such assumptions on characterizing effective networks, and the utility of assumption-free approaches for formalizing CCEP analysis (29, 35, 60). This aligns with our findings; however, to our knowledge, no other study has directly supported this notion with quantitative evidence. To this end, we used a random forest model to assess the predictive strength across a panel of CCEP features for classifying which cingulate subregion was stimulated. This approach revealed the CCEP features with the highest predictive strength for characterizing Cg effective networks. We found that assumption-based waveform features consisting of qualities of N1 and N2 peaks had the lowest predictor strength for classifying the stimulated Cg subregion. This result indicates that the quantitative content represented by these assumption-based features is inferior to other measures (such as coherence or magnitude) in differentiating Cg networks. This further supports previous assertions that formalized, assumption-free methods of analyzing CCEPs will be essential to characterize and compare effective networks (29).

When evaluating the classification accuracy of our model across cortical regions, we hypothesized that Cg “hub” regions, i.e., regions with near identical primary measures of effective connectivity, would have the lowest classification accuracy, especially due to the similarity of how connected these regions are to each Cg subregion. We did not observe this in our results, indicating that the statistical content of features beyond primary measures of effective connectivity is useful for distinguishing CC subregion networks. Instead, hippocampal and amygdala structures had the lowest classification accuracy, demonstrating low distinguishability

between the statistical content extracted from CCEPs in this region. Interestingly, when comparing the frequency of statistical significance between correctly and incorrectly classified CCEPs, we found that while statistical significance was more prevalent in correctly classified responses, only 18-26% of these responses showed statistical significance. This finding may challenge the notion that traditional statistics are sufficient to fully describe effective networks, and instead, multi-modal approaches consisting of a breadth of features will be necessary in characterizing them.

Limitations and Future Directions

Our study included patients with medically refractory epilepsy requiring invasive monitoring via SEEG. It is possible that excessive epileptiform activity could have an impact on the physiology that has been. Regardless, invasively monitored patients with epilepsy have remained a strong model of normal human physiology (26).

Our study provides a comprehensive profile of the Cg effective connectivity networks. These profiles were constructed using a divergent model of effective connectivity, meaning that the characteristics of the features we extracted reflects the connections from the Cg subregion to regions of the cortex. Since effective networks consist of complex, often multidirectional neural interactions, our results do not reflect a complete model of the human Cg effective networks. To form a complete model of the human cingulate cortex's effective connectivity, future convergent models following similar frameworks will be necessary to characterize all directional patterns of neural interactions (32).

Further, "one-to-all" (divergent) and "all-to-one" (convergent) studies produce data that are often high-dimensional, making interpretation of specific connections difficult. While our present work provides a profile of Cg effective connectivity indexing across all available regions, we do not present detailed qualities of single connections. We believe that future work, centered on single or a few effective pathways, will be essential to approaching a comprehensive understanding of the Cg's effective networks. However, our work provides a comprehensive, methodologically sound profile that we believe will be useful to any future work evaluating Cg connectivity or function.

Another limitation of this study is contact coverage. Invasive studies of human effective connectivity are limited by several factors, including biases in contact placement due to clinical needs. While our study addresses other confounding factors, the resolution of our Cg networks is limited by the placement and density of contacts. The subregions of the Cg consist of parcellations more granular than the level of resolution presented in this study. However, as the

formalization of CCEP analysis and effective network modeling improves within the field, we will be able to produce effective connectivity models with higher resolution and investigate the profile of the Cg in greater detail.

Methods

Participants and data collection: This study includes 15 patients with intractable epilepsy undergoing invasive SEEG monitoring at the Barnes-Jewish Hospital epilepsy monitoring unit (Saint Louis, MO). Patient electrophysiology was recorded using a 256-channel clinical system at a sampling rate of 2 kHz. Data collection and event logging were managed using BCI2000 (61). SEEG electrodes were composed of 4-18 contacts with 1.5-mm insulation between them. Patients included in this study had at least one electrode in the PCC, MCC, or ACC. Patients with major lesions or prior resections were not included in this study. Patients unable to provide informed consent or who refused to give informed consent were not recruited for this study.

Electrode localization and patient brain segmentation: Patient-specific brain models were generated using pre-operative T1-weighted structural MRI, computer tomography scans, FreeSurfer, and the VERA colocalization software (62). The Destrieux Atlas (2009), which includes a cytoarchitectonic and functional parcellation of the cingulate sulci and gyri, was used for cortical parcellation (63). Electrode localization was performed using VERA, which performs an automated cluster-based detection of electrodes using post-operative CT scans, with manual post-detection alignments and adjustments (62). For the standardization of anatomy and electrode locations, brain models and electrodes were projected to MNI space. Electrodes were labeled based on the anatomical label of parcellation nearest to the electrode coordinate center. Electrodes localized in Cg subregions were selected *a priori* for SPES. Electrode pairs for SPES were then defined as any pair comprising at least one electrode in a Cg subregion.

Intracranial SPES: To evoke cortical responses to stimulation of the Cg subregions across the cortex, bipolar SPES was delivered to each Cg subregion. The current work defines three stimulation conditions: stimulation of the ACC, MCC, or PCC. SPES was delivered at 6 mA at 0.5 Hz, using a 0.2-ms biphasic square pulse. These parameters have previously been validated in both clinical mapping procedures and studies of effective connectivity in humans (26, 27, 64). Stimulation was delivered using a Nihon Kohden MS-120B-K-EEG cortical stimulator in conjunction with Nihon Kohden PE-210AK software (Nihon Kohden, Japan). Single pulses were delivered across each preselected electrode pair (dipole) localized within the Cg for each patient.

SPES was delivered for approximately two minutes for 60 stimulation events (trials). Stimulation events were logged using g.TRIGBOX (g.tec medical engineering, Austria) connected via a 3.5 mm tip-sleeve cable connected to the cortical stimulator.

Data Preprocessing: Processing of electrophysiological data and statistical analysis was conducted in MATLAB environment (Mathworks, Nattick, MA, USA). SEEG data were preprocessed by first removing the stimulation artifact across all trials. The onset of stimulation was determined using the logged stimulation events recorded through the g.TRIGBOX. Stimulation artifacts were removed from the SEEG recordings using two-way interpolation. Channels containing excessive epileptogenic activity were excluded from analysis. For each channel and stimulation event, we replaced the signal within a 15-sample window around stimulation onset with an interpolated signal derived from time-reversed pre- and post-stimulation segments, weighted by complementary ramp functions (65). Next, a 5th-order Butterworth high-pass filter with a 0.5 Hz cutoff frequency was applied to remove slow drift. A 60 Hz line noise and its harmonics at 120 Hz and 180 Hz were removed using a 2nd-order infinite impulse response notch filter.

To improve the spatial specificity of recordings, a small Laplacian re-referencing method was used. The signal at each contact was re-referenced by subtracting the average signal recorded from all contacts within a 5-mm radius. The data were then epoched into 1900-ms trials centered about each stimulation event. Finally, the data were normalized by z-scoring to a 900-ms pre-stimulation window taken from the start of the trial to 50-ms before the stimulation event (to avoid any potential contamination from stimulation events).

For downstream analysis of CCEP temporal dynamics and evoked gamma responses, we extracted spectral features from the re-referenced data, creating spectral time series that were then epoched as described above. For temporal dynamics analysis, we isolated the predominant broadband spectral content of the CCEPs (broadband CCEP) by applying a 3rd-order Butterworth bandpass filter with cutoff frequencies of 5-40 Hz. For evoked gamma responses, we extracted broadband gamma activity using a 4th-order Butterworth bandpass filter with cutoff frequencies of 70-170. Both filtering operations used transfer functions to second-order sections (SOS, via the tf2sos MATLAB function) to improve the numerical stability and reduce quantization errors inherent in higher-order digital filters. Data were then filtered using zero-phase filtering (using the filtfilt MATLAB function), which applies the filter in both forward and reverse directions to minimize phase distortion while effectively doubling the filter order.

Data Processing and Statistical Analysis: For statistical analysis, we used non-parametric tests. To account for multiple comparisons for significance tests conducted on data at the level of individual contacts, we used the Benjamini-Hochberg false discovery rate (FDR) procedure to calculate an Alpha value such that the FDR was < 0.0001 .

Measures of Effective Connectivity: We defined two measures of effective connectivity: response coherence and response magnitude. Response coherence was used as a metric to quantify SPES-evoked changes in LFP morphology at each contact relative to pre-stimulation (Figure 2a-2b). This was done by comparing the distributions of inter-trial pairwise Spearman Correlations between pre- and post-stimulation time windows. The pre-stimulation (henceforth referred to as baseline) window was defined by an 850-ms window from the start of each trial, and the post-stimulation (henceforth referred to as CCEP) window was defined as a 700-ms window following stimulation onset. The Cohen's d between these distributions defined the effect size of effective connectivity as measured by response coherence. A Wilcoxon rank sum test was used to determine which contacts were significantly effectively connected to the stimulated region between these distributions.

Response magnitude was calculated by taking the root mean square (RMS) of the CCEP window for each trial. The response magnitude at each contact was defined as the average RMS across trials. A Wilcoxon signed-rank test was used to compare the trial distributions of CCEP RMS and baseline RMS at each contact and determine which contacts had significant changes in LFP magnitude in response to SPES. For each Cg subregion stimulation condition, effective connectivity to a given brain region (measured by both coherence and magnitude) was computed as the mean value across all contacts in that region that showed a statistical significance ($FDR < 0.0001$). The effective connectivity of each brain region to the Cg subregions was then calculated for both coherence and magnitude by taking the average of all significant contacts within each region.

Relative Cingulate Cortex Connectivity: To characterize the relative connectivity of brain regions to the three Cg subregions, we implemented a triangular geometric mean mapping approach. The connectivity values at each region to each of the three subregions were normalized to create barycentric coordinates, which were then mapped onto an equilateral triangle with vertices representing exclusive connectivity to each subregion. Brain regions were included in this relative connectivity analysis if they had valid connectivity values for all three subregion stimulation conditions. For any region-subregion pairing in which no contacts were significant, we assigned a

connectivity value of zero for this analysis. For visualization, RGB color values were derived using trilinear interpolation across the triangular space, thus assigning each region a color corresponding to its relative connectivity to each of the cingulate cortex subregions. When assigning relative connectivity coordinates to cingulate subregions themselves, self-connectivity was set to zero to visualize the relative connectivity to the other two subregions. Any regions with barycentric coordinates within a central circle whose area was 5% of the equilateral triangle's total area were classified as equally connected to all three subregions.

Spatial Organization of Cg Connectivity: To characterize the spatial organization of effective connectivity across the three Cg subregions, we performed a spatial distribution analysis along three orthogonal anatomical axes: anterior-posterior, superior-inferior, and left-right. For each axis, the respective coordinates from significantly responsive contacts were divided into 50 equally spaced bins. The coherence effect sizes were min-max normalized to compare across stimulation conditions. Mean normalized effect sizes were computed for each bin, creating a spatial profile of connectivity strength along each anatomical axis. To capture smoothed spatial variation in connectivity strength while reducing noise from sparse electrode coverage, a 4th-order polynomial trend line was fitted to the binned mean effect sizes for each condition. The spatial centroid of each connectivity distribution was calculated as the weighted center of mass using the following equation:

$$c = \frac{\sum_{i=1}^n x_i \cdot w_i}{\sum_{i=1}^n w_i}$$

Equation 1. Calculation for centroids of connectivity spatial distributions.

Where c is the centroid's position, x_i is the position along the spatial axis, and w_i is the mean effect size at each bin.

To assess whether the spatial distribution of connectivity effects exhibited spatial clustering, we performed Moran's I spatial autocorrelation analysis on the polynomial-fitted connectivity profiles (66). A spatial weight matrix was constructed with an epsilon of 1E-5 to define spatial adjacency relationships between bins based on their proximity along each anatomical axis. For each stimulation condition and anatomical axis, we computed the observed Moran's I statistic and compared it to a distribution of Moran's I statistics generated from 1,000 random spatial permutations to calculate a p-value for each spatial distribution. A p-value < 0.01 was considered significant for this analysis.

Statistical comparisons of spatial centroids between stimulation conditions were performed using a permutation-based pairwise comparison of observed centroid differences

between conditions. For each pairwise comparison between stimulation conditions, we computed the observed difference in spatial centroids and compared this to a null distribution generated by 1,000 random permutations of condition labels. A p-value < 0.01 was considered significant for this analysis.

To examine hemispheric differences in effective connectivity, contacts were grouped into major class regions and tested for significant hemispheric asymmetries in connectivity using a Wilcoxon rank sum test. This was done for major regions, both contralateral and ipsilateral to the stimulation sites within each stimulation condition.

Hierarchical Clustering of Cg Effective Connectivity Networks: To characterize the morphological relationships of CCEPs across the cortex, and organize these regions into clustered Cg subnetworks, we performed a maximum correlation-aligned inter-channel pairwise coherence analysis, followed by a hierarchical clustering analysis. To reduce computational load, we downsampled each average CCEP waveform by a factor of 3.

To compute inter/intra-regional coherence for each stimulation condition, we performed pairwise lag-corrected Spearman's correlations of trial-averaged CCEP waveforms between all significantly coherent contacts both within and between all regions. To account for differences in onset latency of CCEPs between regions or contacts, and better isolate the morphological similarity between CCEPs, we performed a Spearman's cross-correlation between each time series for both baseline and CCEP windows (note that window indices were also adjusted for the downsampled time series). The maximum Spearman's Rho for each pairwise correlation was used to generate distributions of CCEP morphological coherence between brain regions for both baseline and CCEP windows. A Wilcoxon rank sum test between these distributions was used to determine which regions had significant CCEP morphological similarities.

The resulting data structure from this analysis was a correlation matrix for each stimulation condition, representing the average lag-corrected Spearman's correlation between regions. We next applied a mask to this matrix to remove any non-significant indices or indices where we did not have sufficient data to make inter-regional comparisons. To compare the overall morphological similarity of CCEPs between stimulation conditions, we performed a Wilcoxon rank sum test between the correlation values of each correlation matrix.

We then applied agglomerative hierarchical clustering to these matrices to identify subnetworks with shared CCEP morphology for each stimulation condition. For this analysis, we calculated the distance matrix of each correlation matrix by subtracting each index from one. We then computed

a linkage matrix encoding relationships between regions as a dendrogram, which was used for defining clusters of regions.

To select a distance threshold to cluster regions, we implemented an automated selection algorithm. The algorithm evaluated clustering solutions across distance thresholds (τ) from 0.1 to 0.9 with a $\Delta\tau$ of 0.01 using a composite scoring function that combined natural breaks in dendrogram structure, balance in cluster sizes, and Shannon entropy. Before scoring, for each τ , any clusters containing two or fewer regions were not considered and removed from the analysis. Breakpoint detection was calculated by finding the merge densities across the range of merge heights from the linkage matrix Z :

$$M(h) = \frac{Hist.(Z_{:,3}, n_{bins})}{\max(Hist.(Z_{:,3}, n_{bins}))}$$

Equation 2. Calculation of merge densities across dendrogram heights.

Where M is the normalized merge density as a function of dendrogram height h , and $Hist.$ represents the histogram function with $n_{bins} = 20$. Local minima representing natural breakpoints were identified by detecting sign changes in the first derivative of the function M . The score for the natural break points for each τ was calculated as:

$$S_{breaks}(\tau) = 1 - \frac{\min(|\mathcal{M} - \tau|)}{\max(Z_{:,3})}$$

Equation 3. Calculation for the breakpoint score to represent naturally occurring gaps in tree structure at each τ .

To calculate a score to represent the similarity of cluster sizes, we calculated a coefficient of variation CV at each τ as the ratio of the standard deviation in the number of regions per cluster to the average number of clusters (where τ' represents all threshold values). We then calculated a cluster balance score as:

$$S_{balance}(\tau) = 1 - \frac{CV(\tau) - \min_{\tau'}(CV(\tau'))}{\max_{\tau'}(CV(\tau')) - \min_{\tau'}(CV(\tau')) + \epsilon}$$

Equation 4. Calculation for cluster balance score as a metric to represent similarity in cluster size and variance at each τ .

Where $\epsilon = 10E-16$ to prevent division by zero. Shannon entropy was calculated as:

$$E(\tau) = - \sum_{i=1}^k p_i \log_2(p_i + \epsilon)$$

Equation 5. Calculation for Shannon entropy at each τ , representing the diversity of cluster sizes.

Where $p_i = \frac{|c_i|}{\sum_{j=1}^k |c_j|}$ represents the proportion of regions in cluster c_i . The Shannon entropy score was then calculated as:

$$S_{\text{entropy}}(\tau) = \frac{E(\tau) - \min_{\tau'}(E(\tau'))}{\max_{\tau'}(E(\tau')) - \min_{\tau'}(E(\tau')) + \epsilon}$$

Equation 6. Calculation for Shannon entropy at each τ , representing the distribution of regions across clusters.

To account for the number of valid regions included in the clusters and the overall distribution of tree structure at each τ we used a composite weight calculated from the following:

$$w_{\text{adaptive}}(\tau) = \exp\left(-\frac{(r(\tau) - 0.5)^2}{0.15}\right)$$

Equation 7. Adaptive weight to account for the distribution of merge densities above and below τ , where $r(\tau)$ represents the proportion of merges below τ .

$$I_{\text{retention}} = 0.5 + 0.5 \times (1 - \exp(-\sigma_{\text{tree}}))$$

Equation 8. Asymptotic function that prioritizes region retention when dendrogram merge heights are unevenly spaced, avoiding thresholds that would exclude large proportions of regions.

$$\sigma_{\text{tree}} = \frac{\text{std}\left(\text{diff}\left(\text{sort}(Z_{:,3})\right)\right)}{\text{mean}\left(\text{sort}(Z_{:,3})\right)}$$

Equation 9. Implementation of σ_{tree} : a metric measuring variability in spacing between clustering events in the dendrogram. Higher values represent larger variability in spacing between successive merge heights, and lower scores show evenly spaced merge heights. At higher values, breakpoints in the tree structure are more likely to exclude a larger portion of regions. Here, $\text{diff}\left(\text{sort}(Z_{:,3})\right)$ represents the successive differences of merge heights sorted in ascending order.

$$w_{\text{composite}}(\tau) = w_{\text{adaptive}}(\tau) \times \left(\frac{n_{\text{regions}}(\tau)}{n_{\text{total}}}\right)^{I_{\text{retention}}}$$

Equation 10. Composite weight function incorporating adaptive emphasis on regional retention (i.e., minimizing the number of regions removed from the final clustering structure) and preferential selection of τ values near the median tree height.

The composite optimization function was then calculated as:

$$F(\tau) = w_{\text{composite}}(\tau) \times [2.0 \times S_{\text{breaks}}(\tau) + 1.5 \times S_{\text{balance}}(\tau) + 1.5 \times S_{\text{entropy}}(\tau)]$$

Equation 11. Composite optimization function for each τ that takes into account the breakpoint structure of the dendrogram, the number of regions represented across clusters, and the Shannon entropy of clusters.

The optimal threshold was then calculated as $\arg \max_{\tau} F(\tau)$, such that the resulting number of valid clusters (>2 regions) was greater than 1.

To characterize the overall differences in dendrogram diversity, we calculated distributions of cophenetic distance (i.e., pairwise differences in branch length across the entire tree) for each stimulation condition and compared these distributions using a Wilcoxon rank sum test. The dendrogram diversity was evaluated by calculating the cophenetic entropy of these distributions.

Temporal Dynamics of Cg Effective Connectivity Networks: To characterize the onset latency and duration of CCEPs across stimulation conditions, we applied an optimal detection of changepoints algorithm on the mean absolute value of the Hilbert transformed broadband CCEPs (broadband magnitude), using two changepoints (under the assumption that a CCEP contains a start and end) at each contact. This method identified changepoints in the time series such that the statistical properties of data between changepoints were maximally separated from the statistical properties of data outside of the changepoints (67). For the onset detection of CCEPs (or CCEP latency), we applied this method to the first derivative of the broadband magnitude to identify the onset and offset of rate changes in the LFP. For offset detection of the CCEPs, we used the changepoint detection method on the broadband magnitude and used the second identified changepoint. The duration of the response was calculated as the difference between the offset and onset.

To ensure that changepoints were detected within expected onset/offset ranges, we estimated the spectral smearing introduced to the signal through broadband filtering, then excluded any changepoints detected outside of the spectral smearing range. Spectral smearing was estimated by applying the broadband filter to simulated data containing a single square pulse, then measuring the change in pulse onset before and after filtering. We found that the degree of smearing depended on various factors, including the magnitude and slope of the evoked responses, so we were unable to extract accurate onset times of stimulation; however, through normalization, we were able to capture trends in response onset between regions in each stimulation condition.

Evoked Gamma Responses to SPES: We characterized the evoked broadband gamma responses across regions using the extracted broadband gamma. To characterize increases or decreases in broadband gamma power at each contact, we used a rank-biserial Spearman's correlation between the mean absolute value of the Hilbert transformed broadband gamma signal at each contact, and a dichotomy (i.e., a generated data vector of zeros and ones corresponding to the lengths of the baseline and CCEP windows for each signal) (68). We then took the average Spearman's rho across all contacts within each region to represent the change in broadband gamma across regions for each stimulation condition.

Assessing Feature Importance for Classifying Cg Effective Connectivity Networks: To assess the importance of the effective connectivity metrics extracted in this study to classify which Cg subregion was stimulated, we used Random Forest Models (69). The model inputs included features aggregated across subjects at each contact, and the model was trained to classify contacts into three groups, specifically corresponding to which Cg subregion was stimulated (ACC, MCC, PCC). The input features for each contact (see Supplemental Methods for description and methods for all features not outlined above) included response coherence, baseline coherence, Cohen's d of response coherence, response coherence variance, response magnitude, contact position (X, Y, and Z- each a separate feature), response latency, response duration, peak response magnitude, peak response latency, response offset, peak broadband gamma magnitude, evoked gamma response, gamma response latency, gamma response duration, response onset angle, response offset angle, each of the (up to) three dominant CCEP angles (each a separate feature), onset of each of the three dominant CCEP angles (each a separate feature), N1 peak amplitude, N1 peak prominence, N1 peak width, N2 peak amplitude, N2 peak prominence, N2 peak width, N2 Peak latency. The details for the methodology used to extract response angles and N1/N2 features are described in Supplemental Methods.

All models were trained with a 5/10-fold nested cross-validation in aggregate across all subjects (5-fold for inner folds, 10-fold for outer folds). Within each fold, models were optimized using 200 iterations of Bayesian optimization, and weighted classification was employed during training to address class imbalance (i.e., differences in sample number for each condition). Permutation feature importance was used to identify the strongest predictive features of the output based on the trained models (70). Permutation feature importance assesses the contribution of each feature by measuring the impact on model performance when the values of that feature are randomly permuted. First, baseline model performance was established using the unaltered dataset. Then, one feature is permuted at a time, and the resulting decrease in model accuracy is

recorded. This process is repeated for each feature, with greater drops in performance indicating more important features. For this analysis, permutation importance was calculated across all models trained during cross-validation, and the feature weights for each model were averaged. This process was repeated 1,000 times per model, and the results were averaged to yield a single importance weight for each feature.

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Data and materials availability: All code used to process, analyze, and visualize data is available on GitHub, along with the processed data used in the present study. Raw data will be made available upon reasonable request to the corresponding author, and are subject to a material transfer agreement since they are acquired from human patients.